

Article

Rapeseed Green Manure Coupled with Biochar and Vermicompost Enhances Soil Aggregates and Fungal Communities in Gleyed Paddy Fields

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Abstract

The gleyed paddy soils in subtropical China, characterized by poor structure, high reductive substances, and low fertility, pose challenges to sustainable agriculture. This study investigates the improvement effects of applying rapeseed green manure in combination with biochar or vermicompost through field experiments, aiming to provide a theoretical basis for the organic improvement of gleyed paddy soils. The experiment included four treatments: control (CK), rapeseed green manure (GM), GM + biochar (GMB), and GM + vermicompost (GMVC). Soil physicochemical properties, aggregate stability, and fungal communities were analyzed after rice harvest. GM significantly increased the total nitrogen (TN) content in the 0–10 cm soil layer and decreased the Fe²⁺ and total glomalin-related soil protein (T-GRSP) contents. GMVC further increased the pH value, available potassium (AK) content, and Shannon index in the 0–10 cm soil layer, decreased the available phosphorus (AP) content, and increased the proportion of macro-aggregates (>2000 μm) and decreased the fractal dimension (D) in the 10–20 cm soil layer. Compared with GMVC, GMB more significantly increased the soil organic carbon content and regulated the ratio of EE-GRSP/T-GRSP in the 0–10 cm soil layer. Fungal community analysis showed *Ascomycota* dominance. Pearson analysis showed *Westerdykella* enrichment significantly correlated with reduced T-GRSP. Monte Carlo tests identified pH and SOC as key factors shaping fungal communities. The GMB strategy mitigates reductive stress, enhances nutrient availability, and activates microbial functionality. These findings offer insights and frameworks for sustainable soil management in subtropical rice agroecosystems.

Keywords: gleyed paddy fields; rapeseed green manure; biochar; vermicompost; soil aggregate stability; fungal community



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1. Introduction

Gleyed paddy soils, characterized by prolonged waterlogging and strong reduction conditions [1], are widely distributed in subtropical China, accounting for 15.07% of total paddy fields [2]. These soils exhibit severe constraints including low oxygen availability, high concentrations of reductive ions (e.g., Fe²⁺, Mn²⁺), and poor aggregate stability due to

the collapse of soil structure under anaerobic environments [3]. Such adverse conditions not only restrict rice yield and nutrient availability but also suppress microbial activity, particularly fungal communities that play critical roles in organic matter decomposition and aggregate formation through glomalin secretion [4]. Consequently, improving soil physical structure and reactivating microbial functions are key challenges for sustainable utilization of gleyed paddy systems.

Organic amendments, such as green manure and biochar, have shown potential in ameliorating degraded soils [5]. Rapeseed (*Brassica napus* L.) green manure has strong adaptability and can be used as a pioneer crop to improve problematic soils. During its growth, it can optimize soil structure, increase soil nutrients, and enhance soil fertility [6], while biochar can absorb harmful substances and improve soil physicochemical properties through its porous structure [7]. Vermicompost can enhance soil fertility and stimulate the activity of soil microorganisms [8]. However, current studies predominantly focus on single amendments in well-drained soils, with limited attention to their synergistic effects in gleyed paddy fields where waterlogged conditions fundamentally alter amendment–soil interactions. Most research has focused [9] on red soils [10], sandy black soils [11], and sandy loam soils [12], with limited attention given to gleyed paddy fields. At the same time, most studies have focused on the effects of the application of green manure in combination with biochar or vermicompost on soil fertility and nutrient improvement, as well as on microbial communities, while there is a lack of research on changes in soil aggregate structure [9,13]. Moreover, the mechanisms by which these amendments regulate fungal community composition and glomalin-mediated aggregate stability remain poorly understood, especially in highly reduced soil environments.

Given the unique challenges posed by these soils, there is a pressing need to explore the effects of organic amendments, particularly in combination with green manure, on soil properties and microbial communities in gleyed paddy fields. This study focuses on gleyed paddy fields in the Jiangnan Plain, a region with significant agricultural importance but severe soil limitations. By integrating rapeseed green manure with biochar and vermicompost, we aim to investigate their combined effects on soil physicochemical properties, aggregate stability, and fungal community composition. The specific objectives are to examine (1) how rapeseed green manure coupled with biochar or vermicompost modulates soil redox status and aggregate stability; (2) the response patterns of fungal communities to amendment-induced changes in soil physicochemical properties; (3) the linkage between glomalin dynamics and fungal taxa enrichment. The findings will provide mechanistic insights into improving gleyed paddy soils through targeted organic management strategies, in the hope of providing a theoretical basis and data support for the organic improvement of gleyed paddy soils.

2. Materials and Methods

2.1. Experimental Site Description

The field experimental site for this study was located in the town of Guanyin Tang in Jingzhou City, Hubei Province, China (30°20′14″ N, 112°30′45″ E, altitude: 16 m). This location is situated within the Jiangnan Plain Lake District, characterized by a subtropical monsoon humid climate, with an annual mean temperature of 16.1 °C and mean annual rainfall of 1100 mm. The study area primarily comprises flooded paddy fields utilized for integrated rice–crayfish farming systems. The soil is classified as gleyed paddy soil, originating from lake sediments. Table 1 presents the baseline data regarding the fundamental physicochemical properties of the soil, which were systematically measured in September 2022, prior to the commencement of the experimental procedures.

Table 1. The physical and chemical properties of the tested soil.

Depth (cm)	pH	SOM (g/kg)	AN (mg/kg)	AP (mg/kg)	AK (mg/kg)	BD (g/cm ³)
0–10	6.65	31.81	204.66	9.88	236.19	1.14
10–20	6.59	30.37	206.82	8.26	255.70	1.37

2.2. Experimental Design

The experiment was designed with four treatments: (1) control (CK, conventional practice with chemical fertilizers only); (2) green manure (GM, rapeseed green manure incorporation at 3.0 t·ha^{−1} fresh weight); (3) GM + biochar (GMB, rapeseed green manure combined with rice straw biochar at 9.0 t·ha^{−1}); and (4) GM + vermicompost (GMVC, rapeseed green manure combined with vermicompost at 3.0 t·ha^{−1}). The plot area was 30 m² (5 m × 6 m). A surrounding ditch 40–50 cm deep was dug around the experimental field. During the green manure rapeseed season, the crops were planted in separate ridges. A soil ridge 30 cm wide and 30 cm high was built between each plot. The surroundings were isolated on both sides with a 100 cm wide black plastic film. There was a 2 m interval between replications, and two drainage ditches were made. Each plot had a separate irrigation and drainage system to prevent the intermingling of fertilizer and water. The green manure rapeseed cultivar utilized in the experiment was “Zhongyoufei No. 1” (the rapeseed green manure variety was selected and provided by the Oil Crops Research Institute of the Chinese Academy of Agricultural Sciences, Wuhan, China) with a sowing rate of 11.25 kg·ha^{−1} and a fresh biomass incorporation rate of 3.0 t·ha^{−1}. To ensure uniformity across all rapeseed green manure application plots, excess biomass was removed, and deficiencies were supplemented following the measurement of actual yields in each plot. The moisture content of the green manure was 88.2%, and the dry basis contents of nitrogen, phosphorus, and potassium were 2.05%, 0.27%, and 3.51%, respectively. The rice straw biochar, prepared by pyrolysis of rice straw at around 500 °C under an oxygen-limited condition, was applied at a rate of 9.0 t·ha^{−1}. Analytical characterization revealed that the biochar contained total carbon (C), nitrogen (N), phosphorus (P), and potassium (K) concentrations of 45.0%, 0.22%, 0.06%, and 1.18%, respectively, with a pH value of 9.71. The vermicompost organic fertilizer, produced by Hubei Tianshenjia Bioenvironmental Technology Co., Ltd. (Huanggang, China), was applied at a rate of 3000 kg·ha^{−1}. This vermicompost was derived from cow manure decomposed by *Eisenia fetida* and exhibited total C, N, P, and K contents of 24.67%, 0.88%, 1.15%, and 1.37%, respectively, with a pH of 7.62.

The green manure rapeseed was sown on 22 October 2022, and incorporated into the soil on 11 April 2023. The application of biochar and vermicompost was conducted on 21 May 2023. The test crop was the rice variety “Jufeng You 248” (the rice variety was purchased from the local market and was bred by Anhui Guorui Seed Co., Ltd., Hefei, China) which was transplanted on 23 May 2023, at a planting density of 16.7 cm × 26.7 cm, with 2–3 seedlings per hill. During the rice-growing season, the application rates of chemical fertilizers were the same for all treatments, with the amounts of nitrogen (N), phosphorus (P₂O₅), and potassium (K₂O) being 165 kg·ha^{−1}, 60 kg·ha^{−1}, and 75 kg·ha^{−1}, respectively. Specifically, 50% of the nitrogen fertilizer was applied as a base dressing, 25% as a tillering fertilizer, and 25% as a panicle fertilizer; 50% of the potassium fertilizer was applied as a base dressing and 50% as a panicle fertilizer; all of the phosphorus fertilizer was applied as a base dressing. The types of fertilizers used were urea (containing 46% N), calcium superphosphate (containing 12% P₂O₅), and potassium chloride (containing 60% K₂O). The experimental field was uniformly managed according to local high-yield cultivation practices, including water management and pest and disease control.

2.3. Sample Collection and Measurement

Soil sampling was conducted in September 2023 following the rice harvest. Within each plot, an S-shaped sampling strategy was employed to collect soil samples from two distinct depth intervals: 0–10 cm and 10–20 cm. The collected samples were thoroughly homogenized and subsequently divided into two subsamples. The first subsample was allocated for the analysis of soil physicochemical properties, while the second was preserved at $-80\text{ }^{\circ}\text{C}$ for subsequent extraction of soil microbial DNA. Additionally, intact soil cores were extracted from each plot using a stainless-steel shovel, ensuring samples were collected from both soil layers. After carefully removing visible roots, these intact soil samples were transferred into water pots for the determination of soil aggregate stability and related properties.

The soil physicochemical properties were determined according to a series of standard methods [14]. BD was measured using the weight method; soil pH was determined using a potentiometer method (with a water-to-soil ratio of 5:1); SOC content was measured using the concentrated sulfuric acid-potassium dichromate external heating method; TN content was determined using the Kjeldahl nitrogen determination method; AP content was measured using the $0.5\text{ mol}\cdot\text{L}^{-1}$ NaHCO_3 extraction-colorimetric method; AK content was determined using the $1.0\text{ mol}\cdot\text{L}^{-1}$ ammonium acetate extraction-flame photometry method; ferrous iron was measured using the o-phenanthroline colorimetric method [15].

The soil aggregate classification was conducted employing the wet sieving technique, as outlined by Elliott [16], which categorizes aggregates into four distinct size fractions: (1) macro-aggregates ($>2000\text{ }\mu\text{m}$); (2) meso-aggregates ($250\text{--}2000\text{ }\mu\text{m}$; subsequently, both macro- and meso-aggregates are herein collectively designated as larger aggregates, encompassing all particles $>250\text{ }\mu\text{m}$); (3) micro-aggregates ($53\text{--}250\text{ }\mu\text{m}$); and (4) silt and clay fractions ($<53\text{ }\mu\text{m}$).

The stability of water-stable aggregates was indicated by D :

$$D = 3 - \frac{\log(w_i/w_0)}{\log(\bar{d}_i/\bar{d}_{max})} \times 100$$

In the formula, $\bar{d}_{max}$ represents the average diameter of the largest particle size class, w_0 is the total mass of all particle size classes, w_i is the mass percentage of the i -th particle size class aggregates (%), and $\bar{d}_i$ is the average particle size between the two sieve fractions d_i and d_{i-1} .

The extraction of balloonmycin-associated soil protein (GRSP) was conducted following the established protocol by Wright and Upadhyaya [17]. For the extraction of easily extractable GRSP (EE-GRSP), 1 g of air-dried soil (sieved through a 2 mm mesh) was placed in a centrifuge tube, followed by the addition of 8 mL of $20\text{ mmol}\cdot\text{L}^{-1}$ sodium citrate buffer (pH 7.0). The mixture was subjected to autoclaving at $121\text{ }^{\circ}\text{C}$ for 60 min, subsequently centrifuged at $10,000\times g$ for 6 min, and the supernatant was collected. For total GRSP (T-GRSP) extraction, 1 g of air-dried soil (sieved through a 2 mm mesh) was treated with 8 mL of $50\text{ mmol}\cdot\text{L}^{-1}$ sodium citrate buffer (pH 8.0). The extraction process involved autoclaving at $121\text{ }^{\circ}\text{C}$ for 60 min, centrifugation at $10,000\times g$ for 6 min, and collection of the supernatant, with this procedure repeated five times to ensure complete extraction. Protein quantification was performed using bovine serum albumin as a standard, employing the Coomassie Brilliant Blue protein assay method. The content of difficultly extractable GRSP (DE-GRSP) was calculated as the arithmetic difference between T-GRSP and EE-GRSP.

2.4. Soil DNA Extraction and High-Throughput Sequencing

The extraction of total genomic DNA of soil microbial communities was performed according to the instructions of the E.Z.N.A.[®] Soil DNA Kit (Omega Bio-tek, Norcross, GA, USA). The quality of the extracted genomic DNA was detected by 1% agarose gel electrophoresis, and the concentration and purity of DNA were measured using the FastPure Soil DNA Isolation Kit (YH-Soil, MJYH, Shanghai, China).

Using the extracted DNA as the template, the fungal ITS region was amplified by PCR with the forward primer ITS1F (5'-CTTGGTCATTTAGAGGAAGTAA-3') and the reverse primer ITS2R (5'-GCTGCGTTCTTCATCGATGC-3') [18], both carrying Barcode sequences. The PCR reaction system was as follows: 4 µL of 5 × TransStart FastPfu buffer (TransGen Biotech Co., Ltd., Beijing, China), 2 µL of 2.5 mM dNTPs, 0.8 µL of forward primer (5 µM), 0.8 µL of reverse primer (5 µM), 0.4 µL of TransStart FastPfu DNA polymerase, 10 ng of template DNA, and the total volume was made up to 20 µL. The amplification procedure was as follows: initial denaturation at 95 °C for 3 min, followed by 27 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s, and extension at 72 °C for 30 s, then a final extension at 72 °C for 10 min, and the product was stored at 4 °C (PCR instrument: ABI GeneAmp[®] 9700, Applied Biosystems, Foster City, CA, USA). The PCR product was recovered using 2% agarose gel electrophoresis, purified using the DNA Gel Extraction and Purification Kit (AxyPrepDNA, AXYGEN, Union City, CA, USA), and the recovered product was quantified using Qubit 4.0 (Thermo Fisher Scientific, Waltham, MA, USA).

The purified PCR products were used to construct libraries using the NEXTFLEX Rapid DNA-Seq Kit (Bioo Scientific, Austin, TX, USA), which included the following steps: (1) adapter ligation; (2) magnetic bead selection to remove adapter self-ligation fragments; (3) library template enrichment by PCR amplification; and (4) magnetic bead recovery of PCR products to obtain the final library. Sequencing was performed on the Illumina MiSeq PE300 sequencing platform (Illumina, Inc., San Diego, CA, USA), which enables paired-end sequencing with a read length of 250 base pairs. The raw data have been deposited to the National Center for Biotechnology Information (NCBI) under the BioProject number PRJNA1236250.

2.5. Statistical Analysis

Statistical analyses were performed using a combination of computational tools and software packages. Data summarization was conducted using Microsoft Excel 2023. One-way analysis of variance (ANOVA) and correlation analysis were performed using SPSS 25.0 software. Duncan's multiple range test was used for multiple comparisons of mean values with significance level set at $p < 0.05$. The proportions of soil aggregates of different particle sizes, fractal dimensions, relative abundances of fungal phyla, and Pearson correlations among indicators were visualized using Origin 2024. Redundancy analysis (RDA) was performed to investigate the effects of soil physicochemical properties on soil fungal community structure. A Monte Carlo permutation test (999 permutations) was conducted using R 4.4.1 to assess the significance of soil physicochemical properties influencing soil fungal communities.

3. Results

3.1. Effects of Different Treatments on Soil Physicochemical Properties

The application of rapeseed green manure significantly influenced soil physicochemical properties, with differential effects observed between the 0–10 cm and 10–20 cm soil layers (Table 2). Relative to the control (CK), the green manure treatment (GM) resulted in higher TN content but lower pH and Fe²⁺ concentration in the 0–10 cm soil layer, while increasing Fe²⁺ content in the 10–20 cm soil layer. The combined green manure and biochar

treatment (GMB) exhibited additional beneficial effects on the 0–10 cm soil layer soil properties, demonstrating increased pH, SOC, and AK along with reduced BD. While the GMB treatment maintained lower Fe^{2+} levels than CK in the 0–10 cm soil layer, it enhanced Fe^{2+} accumulation in the 10–20 cm soil layer. The green manure with vermicompost treatment (GMVC) showed distinct modifications compared to the GM treatment alone. The GMVC treatment led to higher pH values in both soil layers, along with increased AK content and decreased AP in the 0–10 cm soil layer. Furthermore, the GMVC treatment reduced Fe^{2+} content in the 10–20 cm soil layer while increasing it in the 0–10 cm soil layer relative to the GM treatment.

Table 2. Effects of different rapeseed green manure coupling on soil physicochemical properties.

Depth (cm)	Treatment	BD (g/cm ³)	pH	SOC (g/kg)	TN (g/kg)	AP (mg/kg)	AK (mg/kg)	Fe ²⁺ (mg/kg)
0–10	CK	1.22 ± 0.03 a	6.42 ± 0.03 a	19.71 ± 0.56 b	2.01 ± 0.06 b	17.90 ± 0.94 ab	366.02 ± 25.02 b	814.45 ± 21.09 a
	GM	1.18 ± 0.05 a	6.26 ± 0.07 b	19.84 ± 0.47 b	2.07 ± 0.04 a	19.89 ± 2.16 a	377.54 ± 11.09 b	594.77 ± 59.32 c
	GMB	1.12 ± 0.02 b	6.44 ± 0.15 a	21.05 ± 0.62 a	2.07 ± 0.00 a	17.91 ± 0.79 ab	416.78 ± 11.97 a	721.16 ± 18.49 b
	GMVC	1.20 ± 0.01 a	6.53 ± 0.11 a	20.00 ± 0.54 b	2.05 ± 0.01 ab	17.63 ± 0.78 b	413.16 ± 10.43 a	685.79 ± 51.35 b
10–20	CK	1.43 ± 0.04 a	6.60 ± 0.08 b	16.19 ± 0.93 a	1.74 ± 0.07 a	12.51 ± 0.75 a	369.67 ± 10.17 a	279.19 ± 8.75 c
	GM	1.40 ± 0.02 a	6.59 ± 0.10 b	16.41 ± 0.78 a	1.78 ± 0.06 a	12.01 ± 0.10 ab	368.21 ± 3.19 a	401.56 ± 38.47 b
	GMB	1.44 ± 0.01 a	6.72 ± 0.05 b	16.51 ± 1.11 a	1.77 ± 0.03 a	12.05 ± 0.41 ab	368.16 ± 9.34 a	595.02 ± 39.04 a
	GMVC	1.48 ± 0.04 a	6.88 ± 0.11 a	16.35 ± 0.93 a	1.71 ± 0.04 a	11.66 ± 0.23 b	356.57 ± 10.02 a	287.52 ± 31.22 c
Treatment		0.000 **	0.000 **	0.180	0.067 *	0.076 *	0.003 **	0.000 **
Depth		0.000 **	0.000 **	0.000 **	0.000 **	0.000 **	0.000 **	0.000 **
Treatment × Depth		0.000 **	0.305	0.463	0.443	0.078 *	0.000 **	0.000 **

The data are the average of four biological replicates; values followed by the same letters are not significantly different and different letters indicate that Duncan's test is significantly different at $p < 0.05$. * indicates a significant level ($p < 0.05$), and ** indicates a highly significant level ($p < 0.01$).

3.2. Effects of Different Treatments on Soil Aggregate Composition and Stability, and GRSP

Soil aggregate distribution was significantly influenced by the treatments (Figure 1). Compared to CK, the GM treatment reduced the mass proportion of aggregates $>2000 \mu\text{m}$ in the 0–10 cm soil layer while increasing the proportion of 53–250 μm aggregates. However, the situation was reversed in the 10–20 cm soil layer. The addition of biochar (GMB) or combined amendments (GMVC) further increased the proportion of macro-aggregates ($>2000 \mu\text{m}$) in the 0–10 cm soil layer while reducing the fraction of micro-aggregates ($<53 \mu\text{m}$) compared to GM alone.

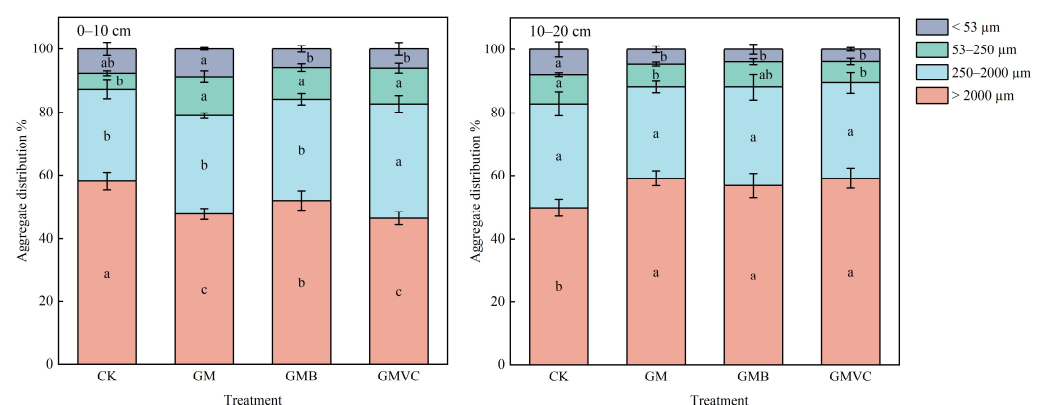


Figure 1. Effect of different treatments on the mass ratio of soil aggregates. The data are the average of four biological replicates; values followed by the same letters are not significantly different and different letters indicate that Duncan's test is significantly different at $p < 0.05$.

Soil fractal dimension was significantly influenced by the treatments (Figure 2). Compared to CK, the GM treatment significantly reduced the fractal dimension (D) in the 0–10 cm soil layer. In the 10–20 cm soil layer, the GMB and GMVC treatments significantly decreased the D compared to CK.

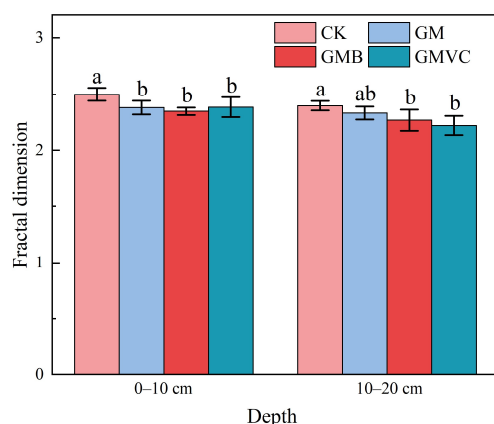


Figure 2. Effect of different treatments on fractal dimension. The data are the average of four biological replicates; values followed by the same letters are not significantly different and different letters indicate that Duncan’s test is significantly different at $p < 0.05$.

Glomalin-related soil protein (GRSP) was significantly affected by the treatments (Table 3), and the interaction between the treatments and soil layer depth significantly affected both DE-GRSP and T-GRSP. Compared to CK, the GM treatment significantly decreased the content of DE-GRSP and T-GRSP in the 10–20 cm soil layer and significantly increased the ratio of EE-GRSP/T-GRSP. Compared to GM, the GMB treatment significantly reduced the content of EE-GRSP and the ratio of EE-GRSP/T-GRSP in the 0–10 cm soil layer, as well as the ratio of EE-GRSP/T-GRSP in the 10–20 cm soil layer, while significantly increasing the content of DE-GRSP and T-GRSP in the 10–20 cm soil layer. The GMVC treatment significantly increased the content of DE-GRSP in the 0–10 cm soil layer and the content of T-GRSP in the 10–20 cm soil layer, and significantly reduced the content of EE-GRSP/T-GRSP in the 0–10 cm soil layer compared to GM.

Table 3. Effects of different green manure coupling organic materials on GRSP in rapeseed green manure.

Depth (cm)	Treatment	EE-GRSP (g/kg)	DE-GRSP (g/kg)	T-GRSP (g/kg)	EE-GRSP/T-GRSP
0–10	CK	8.79 ± 0.89 ab	63.51 ± 1.88 ab	72.31 ± 2.56 a	0.12 ± 0.01 ab
	GM	9.51 ± 0.56 a	62.66 ± 0.66 b	72.18 ± 0.98 a	0.13 ± 0.01 a
	GMB	8.23 ± 0.26 b	64.83 ± 3.10 ab	73.06 ± 3.01 a	0.11 ± 0.01 b
	GMVC	8.49 ± 0.62 ab	66.55 ± 1.58 a	75.04 ± 0.99 a	0.11 ± 0.01 b
10–20	CK	8.22 ± 1.44 a	64.86 ± 3.56 ab	73.08 ± 2.55 ab	0.11 ± 0.02 b
	GM	9.46 ± 0.57 a	56.32 ± 2.46 c	65.78 ± 2.44 c	0.14 ± 0.01 a
	GMB	8.22 ± 0.66 a	67.66 ± 2.68 a	75.89 ± 2.35 a	0.11 ± 0.01 b
	GMVC	8.99 ± 0.24 a	61.04 ± 2.26 bc	70.03 ± 2.15 b	0.13 ± 0.01 ab
Treatment		0.042 *	0.001 **	0.003 **	0.000 **
Depth		0.916	0.065	0.043 *	0.003 **
Treatment × Depth		0.668	0.005 **	0.004 **	0.231

The data are the average of four biological replicates; values followed by the same letters are not significantly different and different letters indicate that Duncan’s test is significantly different at $p < 0.05$. * indicates a significant level ($p < 0.05$), and ** indicates a highly significant level ($p < 0.01$).

3.3. Soil Fungal Community Composition

Significant differences in fungal microbial diversity are presented in Table 4. Compared to CK, the GMVC treatment significantly reduced the Chao 1 index and ACE index in the 10–20 cm soil layer. Compared to the GM treatment, both GMB and GMVC treatments significantly increased the Shannon index in the 0–10 cm soil layer. This indicates that the fungal community richness and diversity in the 0–10 cm soil layer are relatively higher under the GMB treatment, while in the 10–20 cm soil layer, the fungal community richness and diversity are relatively higher under the GM treatment.

Table 4. Effects of organic compounds coupled with green manure in different rapeseed on fungal community stability.

Depth (cm)	Treatment	Chao 1	Shannon	ACE	Simpson
0–10	CK	298.69 ± 93.34 a	3.92 ± 0.51 ab	296.24 ± 89.92 a	0.09 ± 0.08 a
	GM	277.23 ± 68.62 a	3.31 ± 0.63 b	272.75 ± 66.28 a	0.16 ± 0.12 a
	GMB	419.73 ± 75.35 a	4.29 ± 0.08 a	417.61 ± 72.84 a	0.03 ± 0.00 a
	GMVC	300.26 ± 72.09 a	4.28 ± 0.21 a	301.26 ± 72.33 a	0.03 ± 0.01 a
10–20	CK	276.97 ± 107.66 a	4.09 ± 0.06 a	272.55 ± 101.75 a	0.04 ± 0.01 a
	GM	154.56 ± 74.03 ab	3.67 ± 0.44 a	154.23 ± 72.55 ab	0.06 ± 0.02 a
	GMB	146.29 ± 87.27 ab	3.52 ± 0.80 a	144.66 ± 84.98 ab	0.07 ± 0.05 a
	GMVC	111.38 ± 44.13 b	3.02 ± 1.01 a	110.42 ± 43.01 b	0.19 ± 0.21 a

The data are the average of four biological replicates; values followed by the same letters are not significantly different and different letters indicate that Duncan's test is significantly different at $p < 0.05$.

In the 0–10 cm soil layer, after species annotation of the fungal ITS high-throughput sequencing sequences, they were classified into 10 phyla, 31 classes, 55 orders, 106 families, and 166 genera. In the 10–20 cm soil layer, after species annotation of the fungal ITS high-throughput sequencing sequences, they were classified into 11 phyla, 27 classes, 48 orders, 88 families, and 142 genera.

Compared to CK, the GM treatment significantly increased the abundance of *Ascomycota* and *Chytridiomycota* in the 0–10 cm soil layer, and significantly decreased the abundance of *Mortierellomycota*, *Basidiomycota*, and *Rozellomycota* in the 0–10 cm soil layer and the abundance of *Basidiomycota* in the 10–20 cm soil layer. Compared to the GM treatment, the GMB treatment significantly decreased the abundance of *Chytridiomycota* in the 0–10 cm soil layer, and significantly increased the abundance of *Rozellomycota* in the 0–20 cm soil layer. Compared to the GM treatment, the GMVC treatment significantly decreased the abundance of *Ascomycota* and *Chytridiomycota* in the 0–10 cm soil layer and the abundance of *Ascomycota* and *Basidiomycota* in the 10–20 cm soil layer, and significantly increased the abundance of *Mortierellomycota* and *Basidiomycota* in the 0–10 cm soil layer and the abundance of *Mortierellomycota* and *Rozellomycota* in the 10–20 cm soil layer (Figure 3).

3.4. Correlation Between Bacterial Genus and Aggregate Mass Ratio, Average Weight Diameter, and Cemented Substance

RDA analysis was conducted on the fungal community composition in the 0–20 cm soil layer under different treatments (Figure 4). The Monte Carlo permutation test indicated that soil organic carbon (SOC), bulk density (BD), pH, and Fe^{2+} were the main environmental factors causing changes in the fungal community structure in each treatment within the 0–10 cm soil layer, while pH and SOC were the main environmental factors causing changes in the fungal community structure in each treatment within the 10–20 cm soil layer.

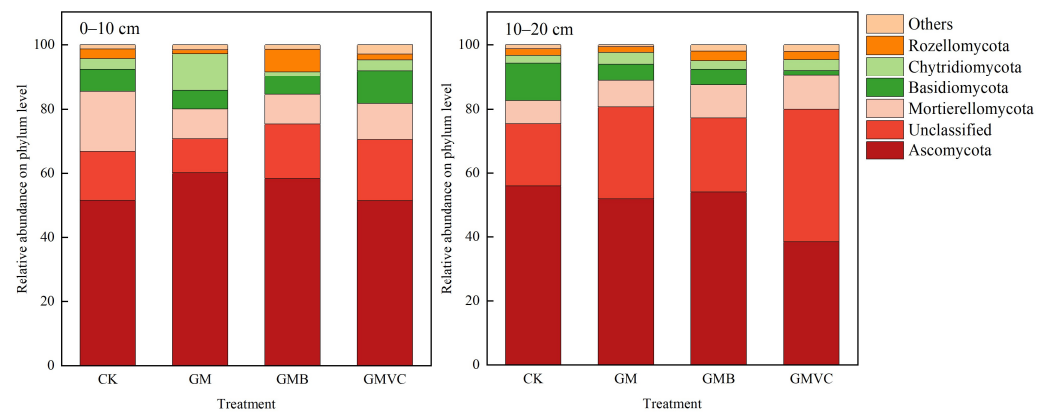


Figure 3. Composition of soil fungal communities in two soil layers with different treatments (phylum).

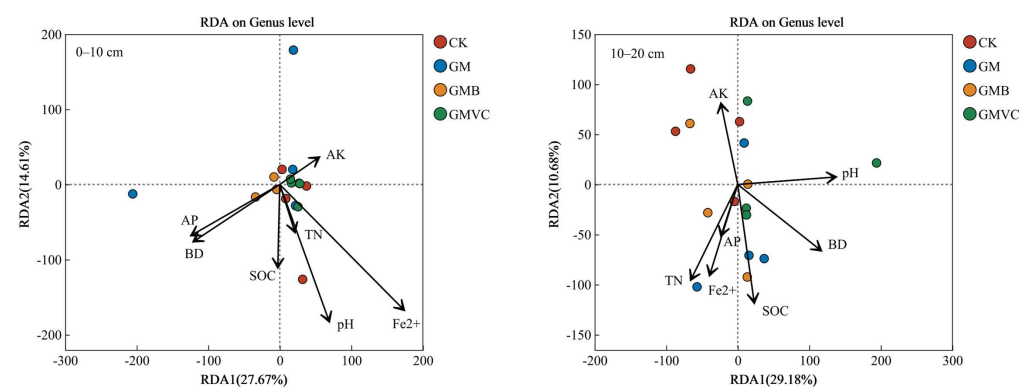


Figure 4. Soil fungal communities (genus level) RDA in different soil layers.

The analysis was conducted at the genus level for the top 10 most abundant genera. In the 0–10 cm soil layer, *unclassified_k_Fungi* was enriched in the GMB and GMVC treatments, *unclassified_c_Sordariomycetes* was enriched in the GMB treatment, and *Trichoderma* and *unclassified_p_Ascomycota* were enriched in the GM and GMVC treatments. In the 10–20 cm soil layer, *unclassified_c_Sordariomycetes* and *Mortierella* were enriched in the CK treatment, *unclassified_o_Sordariales*, *Westerdykella*, and *Scolecobasidium* were enriched in the GM treatment, *unclassified_p_Rozellomycota* was enriched in the GMB and GMVC treatments, and *unclassified_c_Agricomycetes* was enriched in the GMVC treatment (Figure 5A,B).

The results of Pearson correlation showed that there was a significant positive correlation between *Talaromyces* and AK, 53–250 μm aggregate mass ratio and <53 μm aggregate mass ratio in the 10–20 cm soil layer. *Mortierella* was significantly positively correlated with the mass ratio of 53–250 μm aggregates, <53 μm aggregates, and >2000 μm aggregates. *unclassified_c_Rozellomycota* showed a very significant negative correlation; *Scolecobasidium* was significantly positively correlated with TN. *Westerdykella* was negatively correlated with DE-GRSP and T-GRSP. *unclassified_o_Sordariales* was significantly positively correlated with TN. *unclassified_c_Agaricomycetes* was significantly positively correlated with EE-GRSP and negatively correlated with AK. *unclassified_k_Fungi* was significantly positively correlated with BD and pH, and negatively correlated with D (Figure 5C,D).

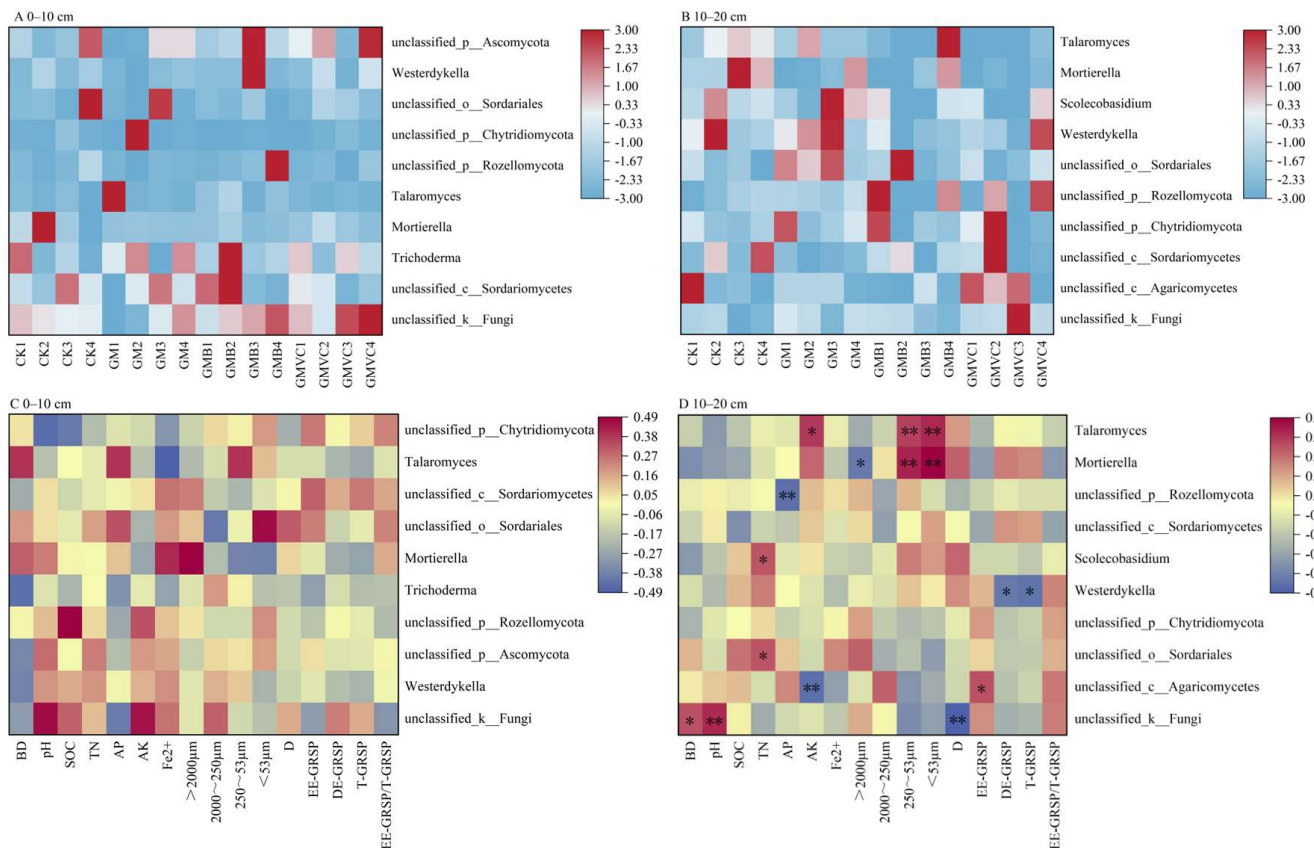


Figure 5. Correlation between horizontal clustering differences of soil fungal genera and soil physicochemical properties, aggregate mass ratio, average weight diameter, cemented material, and differential fungal dominant genera under different treatments. (A,B) are the horizontal heatmaps of fungi, (C,D) is the Pearson correlation heat map of soil layers. * indicates a significant level ($p < 0.05$), and ** indicates a highly significant level ($p < 0.01$).

4. Discussion

4.1. Effects of Rapeseed Green Manure and Coupled Organic Matter Returning on Soil Physicochemical Properties and Aggregate Distribution and Stability

In our study, the interaction between treatment and depth has a significant effect on BD, AP, AK, and Fe²⁺. The nutrient content in soil decreases with increasing depth. Meanwhile, the surface soil is richer in cations than the deeper soil, and thus has a higher content of Fe²⁺. Returning rapeseed green manure to the field can loosen the surface soil and increase soil nutrients [19]. In addition to these benefits, biochar and vermicompost castings can also adsorb heavy metals [20,21].

The incorporation of rapeseed green manure, either independently or in conjunction with organic amendments, exerts a significant influence on soil physicochemical properties, aggregate distribution, and stability. During the decomposition process, rapeseed green manure releases humus, which contributes to the enhancement of soil nutrient availability [22]. Furthermore, it promotes the accumulation of organic carbon and binding agents, thereby ameliorating soil aggregate structure and stability [23]. In the present study, the application of rapeseed green manure led to a notable increase in TN and AP content within the 0–10 cm soil layer, while significantly reducing soil pH and Fe²⁺ concentration. This phenomenon can be attributed to the release of humus during the decomposition of rapeseed green manure, which enriches soil nutrient levels. Additionally, it has been found that the nitrogen transformation processes in the soil, such as nitrification, consume alkaline substances in the soil while releasing hydrogen ions (H⁺), thereby leading to

soil acidification [24]. Moreover, the organic acids released during the decomposition of rapeseed residues also significantly reduce the soil pH value [25]. In comparison to the sole application of rapeseed green manure, the combination with biochar and vermicompost markedly increased soil pH and AK content in the 0–10 cm layer. This outcome is likely attributable to the enhanced nutrient input associated with the combined materials, as well as the potential promotion of organic acid secretion by rice roots in response to the organic amendments. This process results in elevated soil H^+ concentrations and increased cation exchange capacity [26,27]. Moreover, the co-application of rapeseed green manure with biochar significantly augmented SOC content relative to its combination with vermicompost, primarily due to the inherently higher carbon content of biochar compared to vermicompost.

In this study, the application of rapeseed green manure was observed to significantly reduce the mass proportion of large aggregates in the 0–10 cm soil layer, while concurrently increasing it in the 10–20 cm soil layer. This disparity is likely attributable to the heightened susceptibility of the 0–10 cm soil layer to external disturbances. Relative to the control treatment (without rapeseed green manure), the application of rapeseed green manure, either independently or in combination with biochar and vermicompost, significantly elevated the mass proportion of large soil aggregates in the 10–20 cm soil layer. Previous research has substantiated that green manure application facilitates an increase in the content of large soil aggregates. For instance, Xia [28] demonstrated that rapeseed green manure promotes the transformation of micro-aggregates into macro-aggregates within the soil, while Gao [29] reported that the incorporation of rapeseed enhances the proportion of large soil aggregates in rice–rapeseed rotation systems. The D serves as a critical indicator of soil aggregate stability [30]. In this study, compared to the CK treatment, the application of rapeseed green manure in combination with biochar or vermicompost significantly reduced the fractal dimension D value. This phenomenon is primarily attributed to the fact that the application of rapeseed green manure with biochar or vermicompost increases the content of macro-aggregates, which are negatively correlated with the fractal dimension. Therefore, an increase in the proportion of macro-aggregates leads to a decrease in the D value. Park's research indicates that green manure can enhance the stability of soil aggregates [31]. In the present research, the co-application of biochar and vermicompost demonstrated a positive influence on soil aggregate stability. Yang [32] has highlighted that biochar input significantly improves soil aggregate stability, and Li [33] has shown that vermicompost application fosters the formation of large soil aggregates and enhances aggregate stability. AMF contribute to the stability of water-stable aggregates by mitigating the destruction rate of macro-aggregates and the dispersion rate of micro-aggregates through their mycelial networks and GRSP [34]. In this study, the application of rapeseed green manure significantly increased the ratio of EE-GRSP to T-GRSP. Conversely, its combination with biochar significantly reduced the EE-GRSP/T-GRSP ratio while increasing the T-GRSP content. This can be explained by the fact that the decomposition of green manure enhances soil microbial diversity, thereby promoting GRSP accumulation [35], whereas biochar application elevates T-GRSP content by augmenting soil fungal abundance [36].

4.2. Effects of Rapeseed Green Manure and Coupled Organic Matter Returning on Soil Fungal Community

The structural composition and abundance of soil fungal communities constitute essential determinants for sustaining ecosystem productivity and ecological stability [37,38]. In the present investigation, a significantly elevated Shannon index of soil fungi was observed in treatments incorporating rapeseed green manure with biochar or vermicompost, a phenomenon that can be attributed to the enhanced soil nutrient availability following the decomposition of organic matter, thereby establishing more favorable ecological condi-

tions for fungal proliferation. The fungal communities in vermicompost are significantly different from those in the surrounding soil. Studies have shown that vermicompost are enriched with saprophytic fungi, while ectomycorrhizal fungi are relatively less abundant [39]. This difference may lead to changes in the composition of fungal communities in the soil after the application of vermicompost, thereby reducing the overall abundance of fungal communities. At the same time, due to the soil depth, the 10–20 cm soil layer is less disturbed, resulting in changes that are different from those in the 0–10 cm soil layer [40]. The phylum Ascomycota emerged as the predominant fungal group within the study region. These organisms engage in symbiotic associations with plant root systems, facilitating the transformation of organic nitrogen into inorganic forms and promoting root growth kinetics and nutrient acquisition efficiency through their extensive hyphal networks [41]. Zhong [42] demonstrated that the abundance of *Ascomycota* significantly increased following the incorporation of Chinese milk vetch (*Astragalus sinicus* L.) green manure, while Tarin [43] documented the increase in *Ascomycota* populations after bamboo biochar application in red soil ecosystems. These empirical findings are consistent with the results obtained in this study. Furthermore, this investigation identified soil pH and SOC content as the principal environmental determinants influencing the compositional dynamics of soil fungal communities, aligning with the findings of Deshoux [36]. Their comprehensive meta-analysis of biochar application effects on soil microbial communities revealed that fungal community indices exhibit primary dependence on soil texture, pH levels, and SOC content, thereby substantiating the results of the present research.

In our study, the genus *Mortierella* exhibited significant enrichment in the 10–20 cm soil layer in treatments without green manure incorporation, demonstrating a pivotal role in soil nutrient transformation and plant nutrient acquisition processes [44,45]. However, this enrichment pattern was not observed in treatments involving rapeseed (*Brassica napus* L.) green manure incorporation or its concomitant application with biochar and vermicompost. This phenomenon may be attributed to the relatively high proportional representation yet limited absolute biomass of *Mortierella* in the absence of green manure inputs. Furthermore, the study identified *Westerdykella* as a significant component in the 10–20 cm soil layer, where its presence appeared to facilitate the degradation of TOC [46]. Correlation analyses demonstrated significant negative relationships between *Westerdykella* abundance and the concentrations of both DE-GRSP and T-GRSP, suggesting that its proliferation may contribute to the reduction of these soil protein fractions. Comparatively, treatments incorporating rapeseed green manure exhibited significantly lower AP concentrations in the 10–20 cm soil layer, concomitant with enhanced enrichment of *Rozellomycota*. Correlation analyses revealed a significant negative association between reduced AP levels and *Rozellomycota* enrichment, a finding that aligns with Guan's [47] observations regarding continuous straw incorporation. This phenomenon may be partially explained by the potential dominance of unclassified members (*unclassified_p__Rozellomycota*) within the *Rozellomycota* phylum under specific environmental conditions, potentially influencing system sedimentation efficiency and phosphorus removal capacity.

5. Conclusions

This study demonstrates that the integrated application of rapeseed green manure (GM) with biochar or vermicompost significantly enhances soil quality in gleyed paddy fields. GM significantly increased the TN content in the 0–10 cm soil layer and decreased the Fe^{2+} and T-GRSP contents. GMVC further increased the pH value, AK content, and Shannon index in the 0–10 cm soil layer, decreased the AP content, and increased the proportion of macro-aggregates ($>2000\ \mu\text{m}$) and decreased the D in the 10–20 cm soil layer. Compared with GMVC, GMB more significantly increased the SOC content and regulated

the ratio of EE-GRSP/T-GRSP in the 0–10 cm soil layer. Monte Carlo permutation tests identified pH and SOC as pivotal drivers of fungal community succession, and Pearson correlation analysis found that the enrichment of *Westerdykella* was significantly correlated with the reduction of T-GRSP. Collectively, the synergistic strategy of GM combined with biochar concurrently ameliorates reductive stress, enhances nutrient availability, and activates microbial functionality in gleyed paddy soils. These findings provide both mechanistic insights and practical frameworks for sustainable soil management in subtropical rice agroecosystems.

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Abbreviations

The following abbreviations are used in this manuscript:

SOC	Soil organic carbon
BD	Bulk density
TN	Total nitrogen
AP	Available phosphorus
AK	Available potassium
Fe ²⁺	Ferrous iron
GRSP	Glomalin-related soil protein
T-GRSP	Total glomalin-related soil protein
EE-GRSP	Easily extractable glomalin-related soil protein
DE-GRSP	Difficulty extractable glomalin-related soil protein
D	Fractal dimension
AMF	Arbuscular mycorrhizal fungi
AN	Available nitrogen
SOM	Soil organic matter

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